Calculus I

Computation problems #11

Topics:

- Sequences
- Infinite Series
- Integral Test; Comparison Test
- Ratio Test and Root Test

In Problems 10–11, determine which of the sequences a_n converge, and which diverge? Find the limit of each convergent sequence.

Problem 1.

$$a_n = \frac{3^n}{n^3}.$$

Problem 2.

$$a_n = \frac{(10/11)^n}{(9/10)^n + (11/12)^n}.$$

Problem 3. Assume that a sequence converges, find its limit.

$$2,\ 2+\frac{1}{2},\ 2+\frac{1}{2+\frac{1}{2}},\ 2+\frac{1}{2+\frac{1}{2+\frac{1}{2}}},\cdots$$

Problem 4. Write out the first eight terms of each series to show how the series starts. Then find the sum of the series or show that it diverges.

$$\sum_{n=0}^{\infty}\left(\frac{1}{2^n}+\frac{(-1)^n}{5^n}\right).$$

Problem 5. Find a formula for the n-th partial sum of the series and use it to determine if the series converges or diverges. If a series converges, find its sum.

$$\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right).$$

Problem 6. Use the Integral Test to determine if the series converge or diverge. Be sure to check that the conditions of the Integral Test are satisfied.

$$\sum_{n=1}^{\infty} e^{-2n}.$$

Problem 7. Determine the convergence of the following series. Justify your answer. (Remember that there may be more than one way to determine the series' convergence or divergence.)

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n}.$$

Problem 8. Use the Comparison Test to determine if a series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}+1}{\sqrt{n^2+3}}.$$

Problem 9. Determine the convergence of the following series? Use any method, and give reasons for your answers.

$$\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}\ln n}.$$

In Problems 10–11, use any method to determine if the series converges or diverges. Justify your answer.

Problem 10.

$$\sum_{n=1}^{\infty} \frac{n^{\sqrt{2}}}{2^n}.$$

Problem 11.

$$a_1 = 1, \quad a_{n+1} = \frac{1+\tan^{-1} n}{n} a_n \quad (n \geq 1).$$

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Answer to selected problems

1. ∞
2. 0
3. $1+\sqrt{2}$
4. $\dfrac{17}{6}$
5. $S = 1$
6. & 7. converges
8. & 9. diverges
10. converges
11. diverges.